

7. Integration Techniques

7.1. Integration by Parts

Recall that u -substitution corresponded to “undoing” the chain rule (when taking derivatives). What corresponds to the product rule?

Product Rule:

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

Integrating both sides, and rearranging slightly, we end up with the following

$$\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + f'(x)g(x)$$

$$f(x)g(x) + C = \int f(x)g'(x) \, dx + \int f'(x)g(x) \, dx$$

$$f(x)g(x) - \int f'(x)g(x) \, dx = \int f(x)g'(x) \, dx$$

Relabeling the last line with a few substitutions $u = f(x) \Rightarrow du = f'(x) \, dx$ and $v = g(x) \Rightarrow dv = g'(x) \, dx$ we end up with

Theorem 7.1.1 Integration by Parts

$$\int u \, dv = uv - \int v \, du$$

$$\begin{aligned} u &= \sin(x), & dv &= x \, dx, \\ du &= \cos(x) \, dx, & v &= \int x \, dx \\ & & &= \frac{x^2}{2} \end{aligned}$$

7.1 Learning Objectives

- I understand how integration by parts compares to the product rule.
- I understand when and how to use integration by parts.
- I can choose an appropriate u and v
- I can use integration by parts to reduce integrals.

7.1 Section Summary:

- We learned how to “undo” the product rule, via integration by parts.
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